

Heterogeneous Gateway Planning for Semantic Communication in Industrial Networks

Yuyi Sun, *Member, IEEE*, Fangyuan Xing, *Member, IEEE*, Zhenwei Guo, Huimin Chen, and Shibo He, *Senior Member, IEEE*.

Abstract—Semantic Communication (SemCom) has become a promising approach to transmit the information entropy instead of the raw data, which can increase the transmission efficiency and robustness. SemCom recently attracts more attention on the research of Internet of Things (IoT). Previous work focused more on new SemCom framework for IoT applications or lightweight SemCom models which can be easily deployed at IoT end-devices. However, the adaptation of SemCom in industrial networks still remains to be explored. In this article, we focus on gateway planning for SemCom in industrial networks. Specifically, in order to meet the different requirements of various nodes in industrial networks, we build a heterogeneous architecture of gateway planning for SemCom. Then, the Structure-Similarity-Index-Measure (SSIM) model in SemCom is introduced to illustrate the Signal-to-Noise Ratio (SNR) and compression ratio. Based on this, we build the bridge between the SSIM and the number of gateways to analyze our problem. To apply minimum gateways to cover the heterogeneous nodes and meet the SemCom transmission conditions, we formulate the problem as an NP-hard problem and simplify the problem to find the solution. Greedy-Based Algorithm (GBA) and Heuristic-Deep Q Network (H-DQN) algorithm are designed to solve this problem, and simulations are demonstrated to compare H-DQN, Q-learning, GBA and Branch and Bound (BB). The experimental results show that H-DQN outperforms other algorithms, deploying less gateways and saving over 50% of operation time. Besides, the throughput of our system is $1.3\times$ larger than that of the single coverage system with less gateways.

Index Terms—Gateway planning, semantic communication, industrial network, DQN.

I. INTRODUCTION

TODAY has witnessed the prosperous development of Internet of Things (IoT) in various fields, including smart cities, smart home, and industrial manufacturing. IoT empowers intelligent manufacturing by facilitating the connections among diverse industrial sensors, nodes, and gateways. For example, numerous Radio Frequency IDentification (RFID) tags are attached to the cartons that are transported during the production, delivery, and retail of products [1]. Extensive

express-delivery cars and Unmanned Aerial Vehicle (UAV) shuttle back and forth in the streets of the warehouse [2], optimizing routes and pinpointing target products. Cameras are arranged at every corner of the warehouse to form an intelligent security system, tracking cartons, robots, and employees. As a result of the aforementioned requirements and system architectures, massive smart sensors and nodes form massive-connected, heterogeneous, and fragmented IoT industrial networks.

With the rapid development of the Industrial Internet, industrial data is continuously collected, stored, and transmitted. Massive amounts of industrial data can be used to build a virtual digital twin model [3] corresponding to the physical industrial system, helping achieve digital simulation, monitoring, and optimization of industrial processes. Meanwhile, an efficient, intelligent, and sustainable hybrid multi-service stream industrial ecosystem has been built. However, massive industrial data comes from different sources, with diverse business flows and heterogeneous communication protocols, posing higher requirements for the real-time, reliability, and stability of data transmission. In traditional industrial Ethernet networks [4], nodes transmit data based on competition, resulting in severe conflicts among nodes. As data is generated continuously, the total amount of data becomes enormous. Direct data transmission will result in severe resource waste. Time-Sensitive Networking (TSN) [5] introduces time synchronization, flow scheduling, and fault-tolerant communication to distinguish and transmit data according to the priority requirements. It provides low-latency and high-reliability data transmission services to ensure the real-time performance of data with higher priority. However, newly produced data often has a larger information entropy, and TSN does not distinguish the effective information entropy carried by the transmitted data.

Recently, Semantic Communication (SemCom) [6] has emerged as a promising communication technology that goes beyond the limit of Shannon information theory. In order to improve communication efficiency and accuracy, SemCom aims to enhance the understanding and interpretation of messages by extracting the information of the transmitted data based on different applications. SemCom utilizes various advanced technologies, including natural language processing (NLP), machine learning (ML), and artificial intelligence (AI), to extract the features. These technologies enable machines or software to decode and understand the information of the data, which allows for more accurate information retrieval, data analysis and decision making. Instead of transmitting

Yuyi Sun is with the School of Information Science and Technology, Hangzhou Normal University, Hangzhou, China (e-mail: yuyisun@hznu.edu.cn).

Fangyuan Xing is with the School of Cyber Science and Engineering, Southeast University, Nanjing, China (e-mail: fangyuanxing@seu.edu.cn).

Zhenwei Guo is with the Research Center of Cyber Science and Technology, Hangzhou Innovation Institute of Beihang University, Hangzhou, China (e-mail: zhenweiguo0724@buaa.edu.cn).

Huimin Chen and Shibo He is with the State Key Laboratory of Industrial Control Technology, Zhejiang University, Hangzhou, China (e-mail: bethany-chm@zju.edu.cn, s18he@zju.edu.cn).

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the raw data, SemCom only transmits semantic information, which helps reduce the burden on the network especially for the case with a large number of nodes accessing the network or transmitting information. In industrial networks, considering massive connections of nodes and the real-time requirements of data transmission, a large number of nodes are heterogeneous with different communication protocols, uplink and downlink frequencies, transmission bandwidths, etc. Generally, data transmission cannot be interpretable between different networks, protocols, and transmission technologies. The application of SemCom suffers from several challenges. Firstly, massive nodes in industries need to be covered for stable transmission, which requires robust coverage from gateways. Secondly, due to complicate industrial services, nodes in industries have heterogeneous transmission requirements [7], such as various uplink and downlink frequencies and transmission delays. To meet these requirements, the candidate gateways should also be heterogeneous, which brings multiple and non-linear constraints for performance formulation and solving.

In this article, we elaborate the deployment of gateways when applying SemCom in industrial networks. Considering the heterogeneous requirements, we build a new heterogeneous architecture with minimum cost and evaluate the system performance. Specifically, we first model the system with the different cover requirements of types of nodes, and we analyze the performance indicators for data transmission in SemCom to build the relation between SemCom and gateway planning. Secondly, we formulate the problem as an NP-hard problem and simplify the problem based on some physical conditions. Greedy-based algorithm (GBA) and Heuristic algorithm based Deep Q-Network (H-DQN) are proposed to solve this problem. Finally, we evaluate the system performance by comparing four algorithms which includes Branch and Bound (BB) algorithm, GBA, Q-learning algorithm, and H-DQN. The results show that the proposed H-DQN algorithm outperforms the other algorithms, and our system deploys less gateways and obtains higher throughput than that of the single coverage system.

To the best of our knowledge, this is the first work that analyzes the gateway planning problem for SemCom applied in industrial networks. The main contribution of this article is fourfold.

- We build a new heterogeneous architecture of gateway planning for SemCom in industrial networks, which can meet various data transmission requirements of different types of nodes.
- We introduce the Structure-Similarity-Index-Measure (SSIM) model in SemCom, and combine the compression ratio and Signal-to-Noise Ratio (SNR) to build the relation between the gateway architecture and SemCom models.
- The minimum gateway problem is formulated as an NP-hard problem, and GBA and H-DQN are designed to solve this problem. We build the specific reward function in H-DQN, which helps search the minimum gateways faster in this problem.
- Four algorithms with changing parameters are adopted

to investigate the performance, the results of which show that the proposed H-DQN outperforms the other algorithms with less gateways and over 50% of operation time saving. The experimental results also show that the throughput of our system is $1.3\times$ larger than that of the single coverage system.

The remainder of this article is organized as follows. In Section II, we introduce the related work on SemCom. Section III proposes the heterogeneous coverage modeling for SemCom in industrial networks, including the architecture of the heterogeneous network, and the transmission models in SemCom. Section IV formulates the original problem and simplifies the problem. GBA and H-DQN are designed in Section V and evaluated in Section VI. Finally, Section VII concludes this article.

II. RELATED WORK

A. SemCom Based on Different Objects

Some existing studies have designed SemCom network framework based on different data transmission objects. Among them, for text data, DeepSC [8] was a SemCom framework based on deep learning, which employed DNN model to design encoders and decoders, and optimized the network framework gradually by building semantic similarity as an evaluation criterion. For voice data, DeepSC-S [9] introduced an attention mechanism to improve the accuracy of voice data restoration. A semantic slicing model SeSM [10] was designed for image data transmission, and a multi-layer network architecture was proposed to enable transmission tasks to adapt to dynamic environmental changes as soon as possible. Emotion-SemCom [11] proposed a deep neural network based on Transformer and leveraged an external knowledge base to extract emotional knowledge at the transmitter, and employed an emotion-semantic knowledge at the receiver for semantic recovery. A SemCom model [12] was proposed to improve the framework of intelligent transportation.

B. SemCom Architecture for IoT

In traditional IoT architectures, nodes and sensors in the sense layer collect corresponding IoT information, and the network layer guarantees the data transmission of the nodes, enabling the data collected by various IoT nodes and sensors to be transmitted more reasonably and efficiently to the upper layer for intelligent applications. After applying SemCom, the system requires better real-time performance and a larger volume of transmission data. Some existing studies have proposed new IoT architectures based on SemCom. A new type of SemCom framework consisting of semantic physical coding layer, semantic network protocol layer, and semantic intelligent application layer was proposed [13]. Then, it was proved that this new SemCom architecture could make the Industrial IoT (IIoT) transmission more efficient. A SemCom architecture and the theoretical model of task-based communication system was build in [14], and the importance of Joint Source-Channel Coding (JSCC) and Age of Information (AoI) in the SemCom architecture was analyzed. A SemCom architecture [15] was proposed for intelligent transportation.

A SemCom architecture that treated IoT devices as users was designed in [16], where the semantic information was shared among the knowledge bases. The challenges and key technologies faced by SemCom enabling 6G was discussed [17], as well as the data flow of semantic information in IoT architectures.

Considering the IoT devices are always low-memory and power-hungry, some research has begun to focus on lightweight SemCom network architecture suitable for IoT devices. ADNET was used in [18] and the authors designed a sparsification model based on attention-guided denoising convolutional neural network. Besides, the model makes it more suitable for distributed devices in IoT by reducing the number of parameters and precision. DeLighT [19] adopted Facebook's lightweight Transformer model to design a lightweight JSCC method, which reduced the operation time of the SemCom model. ConvNets-based [20] SemCom framework for IoT was proposed, and the authors collected real hardware sensing data to verify that the proposed framework could truly save certain network resources.

C. Resource Allocation in SemCom

Resource allocation is a prosperous topic in SemCom, which has attracted much attention from academia. The authors proposed a dynamic multiplexing and co-scheduling scheme to accommodate both semantic and URLLC traffic [21]. Simulation results demonstrated that the proposed method effectively enhanced the quality of semantic services and satisfied the latency requirements of URLLC traffic. The challenge of accommodating multiple users in a delay-intolerant system was addressed [22], where higher compression ratios saved channel resources but might cause semantic distortion, and lower ratios demanded more resources. To tackle this, the authors optimized both the compression ratio and resource allocation, and designed Compression Ratio and Resource Allocation (CRRA) algorithm and CRRAUS, which reduced the size of transmitted data by up to 80% without compromising the task success probability. YOLOv7-X [23] was adopted to design a SemCom framework to construct a virtual apple orchard, which reduced transmission costs while enhancing the transmission quality of important information in communication services and contributed to the efficient management of digital twins. Other research focused on formulating optimized problem based on some system parameters, such as number of nodes [24], energy consumption [25], Quality of Experience (QoE) [26], delay [27], throughput [28], or building a ubiquitous SemCom optimization objectives [29].

Random access is another prosperous area in resource allocation for SemCom. Theoretical model [30] and simulation results [31] were discussed to elaborate the feasibility of random access based on SemCom. A system based on JSCC via a non-orthogonal generalization of Type-Based Multiple Access (TBMA) [32] reduced error rate efficiently. Based on the shared codebook [33], random access model was built to improve the system performance.

Given the features of data transmission in IoT devices, current research focused on designing different network architectures to optimize task objectives and build lightweight

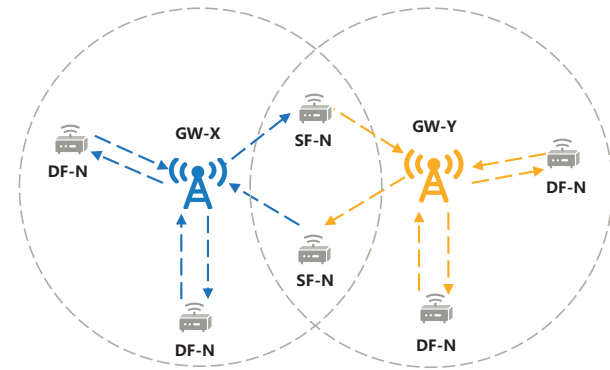


Fig. 1: The heterogeneous coverage model. (SF-Ns: same frequency nodes, DF-Ns: different frequency nodes, GW-X: type X gateways, GW-Y: type Y gateways.)

or sparse network models. However, the importance of gateway coverage in industrial networks and its impact on the performance of SemCom remain to be explored. In SemCom architectures for real-time and hybrid streams in industrial scenarios, the amount of relevant information obtained by IoT nodes from sensing data also needs to be transmitted through the SemCom system, such as the frequency of cameras, the location of deployed sensors, and the information newly created. The aforementioned changes pose new constraints and challenges for the coverage and deployment of IoT edge gateways based on SemCom. Therefore, it is significant to design efficient and economical approaches to deploy gateways with limited resources.

III. HETEROGENEOUS COVERAGE MODELING FOR SEMCOM IN INDUSTRIAL NETWORKS

In this section, we first introduce a heterogeneous coverage model in industrial networks, and then, we propose several basic system models for building SemCom in industrial networks.

A. Heterogeneous Coverage Modeling

Since industrial networks are always massive-connection and hybrid, we build a heterogeneous model of gateway planning. In a heterogeneous hybrid network, we have several types of nodes to be covered, such as the nodes that use different modes for uplink and downlink. We name the nodes that use the same frequency for uplink and downlink as SF-Ns (Same Frequency-Nodes), and the nodes that use different frequencies for uplink and downlink as DF-Ns (Different Frequency-Nodes). In order to guarantee the communication of the nodes, we also need heterogeneous types of gateways to cover the nodes. The heterogeneous gateways are deployed with opposite frequencies for uplink and downlink to meet the requirements of the nodes. Therefore, different kinds of nodes should be allocated with various coverage strategies. A DF-N should be covered by one type of gateway that has consistent frequencies for uplink and downlink. An SF-N should be covered by at least two types of gateways, one type of which communicates with the SF-N for uplink and the other

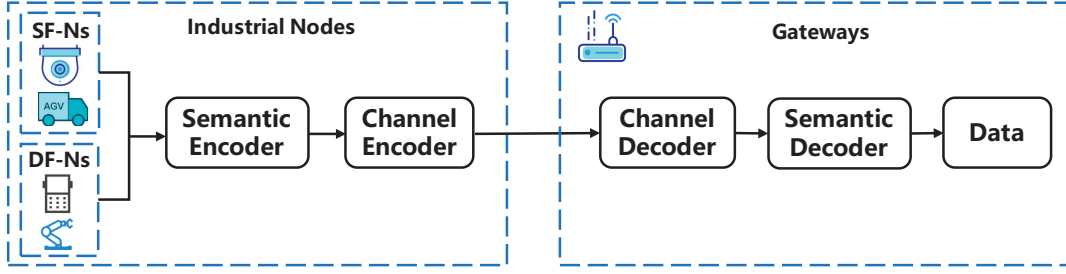


Fig. 2: The architecture of SemCom in industrial networks.

one for downlink. Meanwhile, these gateways should share the same knowledge base in order to validate the uplink and downlink information of the nodes. When applied SemCom, the SemCom encoder and the decoder should be deployed in the gateways and the nodes. Each node is allocated with a special industrial task, such as transmitting the digital twin data of a mechanical arm, transmitting the volume data of some corrugated paper boxes. In Fig. 1, it is assumed that we have some DF-Ns and SF-Ns and two gateways in this network. We define one of the gateways as GW-X, the other as GW-Y. GW-X is allocated with frequency f_1 for uplink and f_2 for downlink. GW-Y is allocated with frequency f_2 for uplink and f_1 for downlink. The DF-Ns use f_1 for uplink and f_2 for downlink or f_2 for uplink and f_1 for downlink, while the SF-Ns use f_1 both for uplink and downlink or f_2 both for uplink and downlink. As Fig. 1 shows, the DF-Ns only communicate with GW-X or GW-Y based on f_1 or f_2 for uplink or downlink, and meanwhile, the SF-Ns communicate with both GW-X and GW-Y for uplink or downlink. That is to say, the DF-Ns only need to be covered by GW-X or GW-Y, and the SF-Ns need to be covered by both GW-X and GW-Y simultaneously.

B. SemCom models

We assume that the SF-Ns and DF-Ns nodes transmit the semantic information based on JSCC strategy. Depending on different downstream industrial tasks, the semantic models extract the features of the raw data and compress it based on different compression settings. The compression ratio defines the transmitted data size and the efficiency of the SemCom network. After this, JSCC network encodes the compressed data based on channel encoder and transmits the data to edge gateways. By decoding the received data, the gateways can partly obtain the raw data. Fig. 2 shows the architecture of SemCom for this heterogeneous network.

SSIM is often adopted to evaluate if the images can be reconstructed accurately. Imagine we deploy IoT nodes to monitor the mechanical arm and build the digital twin in edge gateways, the nodes may be cameras to capture the actions and the degrees of freedom. Generally, our model can be flexibly applied to text and voice transmission. We also adopt SSIM as the criteria to validate the performance of this SemCom network. When the signal is transmitted, the relation between SSIM and SNR is shown in Eq. (1), where $SSIM_k$ represents the SSIM of the node k , and SNR_k is the SNR of the node

k . The values of A_{1,c_k} , A_{2,c_k} , C_{1,c_k} and C_{2,c_k} depend on the setting of compression ratio.

$$SSIM_k(c_k, SNR_k) = A_{1,c_k} + \frac{A_{2,c_k} - A_{1,c_k}}{1 + e^{-(C_{1,c_k} SNR_k + C_{2,c_k})}}. \quad (1)$$

Eq. (2) is the SNR model when the nodes propagate signal.

$$SNR_{ij} = \frac{P_{ij} |g_{ij}|^2 a(f_c, d_{ij})}{d_{ij}^\alpha \sigma^2}, \quad (2)$$

where P_{ij} is the maximum transmit power of the node j when it is covered by gateway i , and g_{ij} is the gain of the channel between the node j and the gateway i . f_c represents the center frequency. d_{ij} is the distance between the node j and the gateway i . α represents the path loss index, and σ^2 is the function of receiver Noise Figure. $a(f, d_{ij})$ is the path loss attenuation function which is defined by Friis transmission equation shown in Eq. (3),

$$a(f_c, d_{ij}) = \left(\frac{v}{4\pi f_c d_{ij}} \right)^2, \quad (3)$$

where c is the velocity of electromagnetic wave.

We introduce Eq. (1)-(3) to formulate the problem of gateway planning for SemCom in industrial networks.

IV. MINIMUM GATEWAY PROBLEM

A. Original Problem

In order to cover the heterogeneous nodes mentioned in Sec. III-A, we separate the SF-Ns into SF-X and SF-Y nodes based on the uplink and downlink frequency. We also divide the DF-Ns into DF-X and DF-Y nodes. There are totally N nodes in this SemCom network, and the number of SF-X, SF-Y, DF-X and DF-Y nodes are respectively n_1 , n_2 , n_3 , and n_4 . That is to say, the sum of n_1 , n_2 , n_3 , and n_4 equals N . We define the total set of the all the nodes as $\mathcal{N}$. There are M candidate locations to deploy gateways, we need to decide which location to deploy gateways and which type of gateway should be deployed. We define the total set of the candidate locations as $\mathcal{M}$. In order to meet the requirements of various nodes and minimize the costs, less gateways should be decided to deploy at candidate locations. We define $\mathbf{X}$ and $\mathbf{Y}$ as $1 \times M$ matrices, x_i and y_i are the element of $\mathbf{X}$ and $\mathbf{Y}$, respectively. $x_i = 1$ means that a gateway of type X is deployed at location i , and $x_i = 0$ represents that no gateway

TABLE I: Notations in this problem

Name	Symbol
M	Number of candidate location of gateways
$\mathcal{M}$	Set of candidate locations
N	Number of the nodes
n_1	Number of SF-X nodes
n_2	Number of SF-Y nodes
n_3	Number of DF-X nodes
n_4	Number of DF-Y nodes
N_1	$n_1 + n_2$
N_2	$n_1 + n_2 + n_3$
$\mathbf{X}$	Matrix describes if GW-Xs is deployed at the locations
$\mathbf{Y}$	Matrix describes if GW-Ys is deployed at the locations
x_i	Elements in $\mathbf{X}$
y_i	Elements in $\mathbf{Y}$
k_{ij}	If node j can be covered by gateway i
P_{ij}	Power of gateway i occupied by node j
P_{max}	Max power of the gateways
$SSIM_{ij}$	The SSIM between node j and gateway i
SNR_{ij}	The SNR between node j and gateway i
g_{ij}	The gain of the channel between node j and gateway i
f_c	The center frequency
d_{ij}	The distance between node j and gateway i
$a(f_c, d_{ij})$	The path loss attenuation function
$A_{1,c_i}, A_{2,c_i}, C_{1,c_i}, C_{2,c_i}$	The constants in the model of SSIM
c_i	The selected compression ratio
v	The velocity of light
P_{ij}^*	Transmit power of gateway i occupied by node j

of type X is deployed at location i . $y_i = 1$ means that a gateway of type Y is deployed at location i , otherwise, $y_i = 0$. k_{ij} is defined as a matrix which describes whether a node can be covered by a gateway or not. We formulate our problem as $\mathcal{P}_1$.

Eq. (5a) and Eq. (5b) mean that the SF-Ns must be simultaneously covered by the type of X and Y gateways. $x_i = 1$ and $k_{ij} = 1$ mean that the gateway i is activated as type X and the node j can be covered by gateway i . $x_i = 0$ means that the gateway i is not activated. If $y_i = 1$ and $k_{ij} = 1$, the gateway i is activated as type Y and the node j can be covered by gateway i . Otherwise, $y_i = 0$ means that the gateway i is not activated. $k_{ij} = 1$ represents that the node j is within the cover range of gateway i , which also means that the gateway i can meet the coverage requirement of node j . Otherwise, $k_{ij} = 0$ means that the node j is out of the cover range of gateway i . As shown in Eq. (5o), x_i , y_i , and k_{ij} are 0-1 variables.

$$\mathcal{P}_1 : \min \sum_{i=1}^M (x_i + y_i). \quad (4)$$

$$\text{s.t.} \quad \sum_{i=1}^M x_i k_{ij} \geq 1, \text{ for each } j, j \in [1, N_1], \quad (5a)$$

$$\sum_{i=1}^M y_i k_{ij} \geq 1, \text{ for each } j, j \in [1, N_1], \quad (5b)$$

$$\sum_{i=1}^M x_i k_{ij} \geq 1, \text{ for each } j, j \in [N_1 + 1, N_2], \quad (5c)$$

$$\sum_{i=1}^M y_i k_{ij} \geq 0, \text{ for each } j, j \in [N_1 + 1, N_2], \quad (5d)$$

$$\sum_{i=1}^M x_i k_{ij} \geq 0, \text{ for each } j, j \in [N_2 + 1, N], \quad (5e)$$

$$\sum_{i=1}^M y_i k_{ij} \geq 1, \text{ for each } j, j \in [N_2 + 1, N], \quad (5f)$$

$$\sum_{j=1}^{N_1} x_i k_{ij} P_{ij} + \sum_{j=N_1+1}^{N_2} x_i k_{ij} P_{ij} \leq P_{max}, \quad (5g)$$

$$\sum_{j=1}^{N_1} y_i k_{ij} P_{ij} + \sum_{j=N_2+1}^N y_i k_{ij} P_{ij} \leq P_{max}, \quad (5h)$$

$$P_{ij} = \frac{SNR_{ij} \cdot d_{ij} \sigma^2}{|g_{ij}|^2 a(f_c, d_{ij})}, \quad (5i)$$

$$a(f_c, d_{ij}) = \left(\frac{c}{4\pi f_c d_{ij}} \right)^2, \quad (5j)$$

$$SSIM_{ij} = A_{1,c_i} + \frac{A_{2,c_i} - A_{1,c_i}}{1 + e^{-(C_{1,c_i} SNR_{ij} + C_{2,c_i})}} \geq \eta_0, \quad (5k)$$

$$n_1 + n_2 = N_1, \quad (5l)$$

$$n_1 + n_2 + n_3 = N_2, \quad (5m)$$

$$n_1 + n_2 + n_3 + n_4 = N, \quad (5n)$$

$$x_i \in \{0, 1\}, y_i \in \{0, 1\}, k_{ij} \in \{0, 1\}, \quad (5o)$$

$$x_i + y_i \leq 1, \quad (5p)$$

$$c_i \in \{c_1, c_2, \dots, c_K\}. \quad (5q)$$

Eq. (5c) and (5d) represent that if the nodes in n_3 can be covered by gateway i of type X, the system can meet the requirements of these nodes. Eq. (5e) and (5f) represent that the nodes in n_4 are covered by gateway i of type Y. Eq. (5g) and (5h) mean that the gateway power occupied by the nodes during data transmission should not be larger than the limited output power of the gateways. Eq. (5i) and (5j) draw the relation between the transmit power P_{ij} and SNR_{ij} . Eq. (5k) represents that the SSIM should not be lower than the threshold η_0 which is the accuracy requirement of the system. Eq. (5l) to (5n) are the total number constraints of the four kinds of nodes. It is noted that once a gateway is deployed at a specific location, no more gateways can be deployed at this location. Therefore, as Eq. (5p) shows, x_i and y_i cannot

be one simultaneously. Eq. (5q) defines the compression ratio that should be selected in the setting set, which has K values of compression ratio. Table I enumerates the notations used in this article.

It is noted that the problem formulated above is non convex and hard to find an appropriate solution. Eq. (5g) to (5k) are extremely nonlinear constraints with dynamic parameters. Therefore, we analyze the problem insights in Sec. IV-B and we simplify the constraints based on some physical situations and reasonable analysis in Sec. IV-C, making problem $\mathcal{P}_1$ explainable and finding feasible solutions by algorithm design.

B. Problem Insights

It is noted that $\mathcal{P}_1$ is a mixed integer linear constraint programming problem after the analysis above. In this subsection, we investigate that the problem of minimum gateways in $\mathcal{P}_1$ is NP-hard. Then, we discuss some insights of this problem.

Theorem 1. *The problem of minimum gateways in $\mathcal{P}_1$ is NP-hard.*

Proof. Given a large number of nodes with location coordinates and some candidate gateway locations also with location coordinates, in order to prove that the problem is NP-hard, we first introduce set cover problem.

Definition of the Set Cover Problem: Given a set $\mathcal{U}$ and a set $\mathcal{S}$ that contains n subsets whose union is the set $\mathcal{U}$. In set cover problem, the goal is to search the minimum subset of $\mathcal{S}$ to cover the set $\mathcal{U}$, such that their union is equal to the set $\mathcal{U}$. Besides, we have weighted set cover problem. Subsets in $\mathcal{S}$ is applied with weights $\lambda_1, \lambda_2, \dots, \lambda_n \in [0, 1]$. Each subset is assigned with different weights, which determines the final cover performance.

In our problem, let the total heterogeneous nodes be the set $\mathcal{U}$, and the set of candidate locations to deploy gateways is $\mathcal{S}$. The goal of the problem is to find minimum subset of the candidate locations of gateways to cover the large number of nodes. Based on different cover requirements of these nodes, we can separate these nodes into two categories and decompose the problem into two sub-problems. As for the SF-Ns, they should be covered by two types of the gateways, which means that we need to find the minimum subsets of the two types of gateways to cover SF-Ns. Thus, the sub-problem to cover SF-Ns is a set cover problem. Analogously, the sub-problems to cover DF-Ns are also regarded as set cover problems.

Based on the other constraints, we need to consider the power, the SSIM, and the SNR threshold when we search the subset to cover the heterogeneous nodes. Thus, the problem is a specific set cover problem to find subset based on complex constraints.

Based on the analysis above, our problem of minimum gateways is a set cover problem. Since set cover problems are NP-hard problems, our problem of minimum gateways is an NP-hard problem, which completes the proof. $\square$

Moreover, it should be noted that the constraints in $\mathcal{P}_1$ are extremely non-linear, dynamic, and uncertain, which makes the problem difficult to find the solution. When we try to find

the subsets of the candidate locations of the gateways, we need to simultaneously consider whether the total transmit power is beyond the limit, whether the SSIM is accurate enough to be decoded, etc. Therefore, before we design algorithms, we try to analyze the elements in these constraints based on some physical situations and simplify the problem.

C. Simplified Problem

In this subsection, we analyze the elements and models in real coverage scenarios, and try to simplify the problem and reduce the complexity of the constraints.

In this problem, we aim to find the smallest number of gateways to cover the nodes, thus, we hope that one single gateway can cover as many nodes as possible. Once a gateway cover a large number of nodes, the gateway power occupied by every node when transmission should be as less as possible. Eq. (2) illustrates the relationship between SNR and the transmit power. Thus, when the transmit power occupied by the node is low, the SNR of the node covered by the gateways should also be low. Eq. (5k) describes the relationship between the SSIM and the SNR, and also draw the threshold requirement of the node transmission in SemCom networks. If SSIM is larger than the threshold η_0 , the signal received by the receiver can be decoded in the semantic decoder. It is noted that when the SSIM is equal to the threshold, the SNR of every nodes can be set at the minimum. Therefore, we can set $SSIM_{ij} = A_{1,c_i} + \frac{A_{2,c_i} - A_{1,c_i}}{1 + e^{-(C_{1,c_i} SNR_{ij} + C_{2,c_i})}} = \eta_0$, and then we can obtain the corresponding SNR based on different compression ratios. When SSIM is fixed, the smaller SNR can be obtained with larger compression ratio. Therefore, by setting the compression ratio and SSIM, SNR can be set at minimum based on these constraints and can be treated as a known variable.

After we obtain the minimum SNR, the uncertain parameter in Eq. (5i) is d_{ij} . By putting the minimum SNR and Eq. (5j) in Eq. (5i), there is a relation between P_{ij} and d_{ij}^3 . As we have already known the locations of the candidates and the nodes, the distance distribution between the candidates and the nodes can be obtained. Thus, $a(f_c, d_{ij})$ can be fixed at a definite value between one single node and one single gateway. By putting Eq. (5j) and (5k) into Eq. (5i), P_{ij} between gateway i and node j can be obtained. By setting the certain P_{ij} between each node and each gateway in Eq. (5g) and (5h), (5g) and (5h) can be transformed to linear constraints. We define the specific transmit power of gateway i occupied by node j as P_{ij}^* . Thus, constraints Eq. (5g) and (5h) can be rewritten as Eq. (6) and (7). Moreover, the constraints can be further simplified. Eq. (5a) and (5c) describe the coverage situation of type X gateways. Thus, we can further combine Eq. (5a) and (5c) as Eq. (8).

$$\sum_{j=1}^{N_1} x_i k_{ij} P_{ij}^* + \sum_{j=N_1+1}^{N_2} x_i k_{ij} P_{ij}^* \leq P_{max},$$

for each $i, i \in [1, M]$. (6)

$$\sum_{j=1}^{N_1} y_i k_{ij} P_{ij}^* + \sum_{j=N_2+1}^N y_i k_{ij} P_{ij}^* \leq P_{max},$$

for each i , $i \in [1, M]$. (7)

$$\sum_{i=1}^M x_i k_{ij} \geq 1, \text{ for each } j, j \in [1, N_2]. \quad (8)$$

We can further combine Eq. (5b) and Eq. (5f) as Eq. (9). It is noted that $x_i \geq 0$, $y_i \geq 0$, $k_{ij} \geq 0$. Thus, Eq. (5d) and Eq. (5e) can be ignored.

$$\sum_{i=1}^M y_i k_{ij} \geq 1, \text{ for each } j, j \in [1, N_1] \cup [N_2 + 1, N]. \quad (9)$$

Therefore, problem $\mathcal{P}_1$ can be transformed to $\mathcal{P}_2$ as follows:

$$\mathcal{P}_2 : \quad \min \quad \sum_{i=1}^M (x_i + y_i). \quad (10)$$

$$\text{s.t.} \quad \text{Eq.(8) - Eq.(9),} \quad (11a)$$

$$\text{Eq.(6) - Eq.(7),} \quad (11b)$$

$$\text{Eq.(5l) - Eq.(5p).} \quad (11c)$$

Theorem 2. Problem $\mathcal{P}_2$ is also NP-hard.

Proof. As the distance distribution and the power distribution are both fixed, it is noted that problem $\mathcal{P}_2$ is a mixed integer linear programming problem. In this problem, we also need to find minimum subset of the candidate locations to cover the nodes, which means that the problem is a set cover problem. Therefore, problem $\mathcal{P}_2$ is also NP-hard. $\square$

Though we can adopt some optimizers like CPLEX or Gurobi to solve the problem, the algorithm complexity is high. Therefore, it is urgent to design efficient algorithms to find the solution.

V. GATEWAY PLANNING SCHEMES

In this section, in order to solve the problem efficiently, we first design GBA to accelerate the procedure. Then, in order to find better solution, we further design H-DQN.

A. Greedy-based Algorithm (GBA)

The goal of the problem formulated before is to search minimum gateways with the definite position of nodes and the candidate position of the gateways to be deployed. We hope that the two types of gateways can cover as many nodes as possible. Since the nodes are supposed to be closer to the target gateway for better communication quality, we first sort the transmit power of the nodes that are within a candidate gateway due to the covering radius. During each iteration, we need to decide whether a gateway should be deployed at the candidate location and which type of gateway (X or Y) should be deployed. Algorithm 1 shows the iteration procedure of GBA. Before the algorithm begins, we initialize the locations of the nodes and the candidate locations of the gateway

randomly. We define a flag matrix to record the covering states of the nodes. Since the SF-Ns should be covered by two type of gateways, the flags of SF-Ns contain two elements a_n and b_n , $n \in [1, N_2]$, while DF-Ns have one element a_n , $n \in [N_2 + 1, N]$. $a_n = 1$ represents the GW-X at candidate location n is activated, otherwise, $a_n = 0$ represents the GW-X at candidate location n is not activated. $b_n = 1$ means that the GW-Y at candidate location n is activated, otherwise, the GW-Y is not activated. When the flag elements of all the nodes are equal to one, the network achieves the goal of coverage. For each iteration, the algorithm should select maximum nodes based on the sorted transmit power of the nodes, and find the index and the type of the corresponding location.

Algorithm 1: The Greedy-based Algorithm

Input: $M, N, k_{ij}, P_{ij}^*, P_{max}$

Output: o

```

1  $o = 0, X_m = 0, Y_m = 0, m \in [1, M];$ 
2 while  $a_n \neq 1$  or  $b_n \neq 1$  for  $n \in [1, N_2]$ ,  $a_n \neq 1$  for
   $n \in [N_2 + 1, N]$  do
3   Select the maximum nodes  $n$  that satisfies  $a_n \neq 1$ 
    or  $b_n \neq 1$  within the range of candidate  $m$ 
    ( $m \in [1, M]$ ) if deploying GW-X and add them to
    a temp set  $C_1$ , and select the nodes within the  $m$ 
    if deploying GW-Y and add them to set  $C_2$ ;
4   Sort the transmit power of nodes in  $C_1$  or  $C_2$  and
    delete the nodes which has larger value that
    exceeds  $P_{max}$ ;
5   Calculate the reserved maximum number of nodes
    that are newly covered by GW-X:  $X_{new}$ ;
6   Calculate the reserved maximum number of nodes
    that are newly covered by GW-Y:  $Y_{new}$ ;
7   if  $X_{new} > Y_{new}$  then
8      $a_n = 1$  and  $b_n = 1, n \in [1, N_2];$ 
9      $X_m = 1;$ 
10  else
11     $a_n = 1, n \in [N_2 + 1, N];$ 
12     $Y_m = 1;$ 
13  end
14  Delete the newly selected candidate location;
15  Delete the newly selected nodes;
16   $o = o + 1.$ 
17 end
18 Return  $o$ 
```

In order to analyze the performance of GBA, the approximation ratio of GBA should be discussed. Since the SF-Ns need to be covered by GW-X and GW-Y simultaneously, the gateway should cover $2i + j$ times if we have i SF-Ns and j DF-Ns in this network. We define $\mathcal{N} = 2i + j$. After running GBA, we obtain the number of the gateway that need to deploy to cover all of the nodes, which is defined as $\mathcal{O}$. We define the optimal solution as $\mathcal{O}^*$.

In each iteration of GBA, the algorithm selects the gateway which can cover maximum nodes that include the SF-Ns and DF-Ns to deploy. If the gateway in each iteration covers the average nodes, the number of iteration would increases,

which results in the worst case of greedy algorithm. This means that one single gateway should cover at least (*Rest nodes / Rest gateways to deploy*) nodes during each iteration. The procedure of how the worst case operates is analysed as follows,

- When GBA begins the first iteration, the chosen gateway should cover at least $\mathcal{N}/\mathcal{O}^*$ nodes. Therefore, there are $\mathcal{N} - \mathcal{N}/\mathcal{O}^*$ nodes left in this network.
- When GBA begins the second iteration, the chosen gateway should cover at least $(\mathcal{N} - \mathcal{N}/\mathcal{O}^*)/(\mathcal{O}^* - 1)$ nodes, and $\mathcal{N} - \mathcal{N}/\mathcal{O}^* - (\mathcal{N} - \mathcal{N}/\mathcal{O}^*)/(\mathcal{O}^* - 1)$ nodes are left.
- ...
- After $\mathcal{N}$ iterations, all of the nodes should be covered.

Therefore, after GBA ends, we have the relation between the optimized solution $\mathcal{O}^*$ and the solution obtained from GBA $\mathcal{O}$, shown in Eq. (12),

$$\left\{ \left(1 - \frac{1}{\mathcal{O}^*}\right) \cdots \left[1 - \frac{1}{\mathcal{O}^* - (\mathcal{O} - 1)}\right] \right\} \cdot \mathcal{N} < 1, \quad (12)$$

Since we have $(1 - \frac{1}{\mathcal{O}^*})^{\mathcal{O}^*} < \frac{1}{e}$, we can obtain Eq. (13),

$$\left(1 - \frac{1}{\mathcal{O}^*}\right)^{\mathcal{O}^* \cdot \ln \mathcal{N}} < \frac{1}{\mathcal{N}}. \quad (13)$$

From Eq. (13), it is noted that when GBA ends, selecting $\mathcal{O}^* \cdot \ln \mathcal{N}$ subsets can cover all the nodes by deploying the candidate gateways. Therefore, GBA has an approximation ratio of $\ln \mathcal{N}$ compared with the optimal solution.

Based on the results above, it is noted that the approximation ratio increases with the increase of the number of the nodes. Though GBA calculates more fast than the optimal solution, the approximation ratio between the two results is still loose. The algorithm still needs to be optimized.

B. Heuristic Deep Q Network (H-DQN) Algorithm

In order to search a more accurate solution, we design the H-DQN. Specifically, we are supposed to define the state, action, and reward function. Besides, it is supposed to define more details in our system models.

Definition 1. If an SF-N is covered by a third gateway or a DF-N is covered by another gateway, the node is a redundant node.

Definition 2. If all the nodes are redundant nodes after the algorithm iteration, the network is a non-redundant network.

In our system model, the iteration ends when all the nodes with different requirements are covered. It is hoped that when the network is a non-redundant network, we can obtain as less number of gateways as possible. Therefore, when we define our state space and action space, we set each node as a redundant node, which means that in each step, we delete the nodes that have already been covered in the last step.

Definition 3. After the gateway covers several nodes in each step, the unoccupied power is defined as e_m , $m \in [1, M]$.

State. The state space contains a $M \times 2$ matrix, the first column is isA_m , which defines whether the candidate gateway is active or not. $isA_m = 1$ means that the gateway at location m is active, otherwise, the gateway is asleep. The second column is e_m , $m \in [1, M]$, as Eq. (14) shows. In each state, the matrix describes the number of the gateways deployed and the maximum nodes that the gateways can cover.

$$state_m = \{isA_m, e_m\}, \quad m \in [1, M]. \quad (14)$$

When a gateway is asleep and the action in one step is to activate this gateway, e_m is equal to the power threshold minus the maximum transmit power of the nodes that the gateway can handle. Shown in Eq. (15), n' is the number of nodes that the m -th gateway can cover. $P_{n'm}^*$ is the power distribution between each node and the M gateways. If a gateway is activate, it cannot be activated or deactivated in next state.

$$e_m = P_{max} - \sum_1^{n'} P_{n'm}^*. \quad (15)$$

Action. The action space is a $1 \times M$ matrix. We should choose an index of the candidate gateway to decide which gateway is to be deployed. We can only decide one gateway to deploy in each action. The action a_m depends on the ϵ -greedy algorithm, which shows in Eq. (16).

$$a_m = \begin{cases} \text{random action}, & \epsilon, \\ \text{argmax } Q(s_t, a_t), & (1 - \epsilon). \end{cases} \quad (16)$$

Reward function. Considering the unoccupied power and the activate state of the gateways change in each step, we propose the reward function based on the two parameters. Combining the condition that the network should be redundant, the gateways should cover as many nodes as possible. Besides, the power of the covered nodes within a single gateway should be as less as possible, which also means that the gateway can cover more nodes. In order to avoid deploying one single gateway for only one or two nodes during the later few steps, we should set an initial cost at the beginning of each step. When the gateway cover more nodes, the accumulated reward can cover the initial cost. Therefore, we set an initial reward to be ready to activate the gateways. Then, during the coverage procedure, based on the action selected by the agent, the algorithm sorts the power and select the nodes to cover. We define the reward function as Eq. (17) shows. $n' > 0$ means that the gateway can cover at least one nodes, while $n' = 0$ means that there is no nodes within the range or the power cannot satisfy the condition. α is the initial reward to activate the gateway. β is the conversion factor to balance the final reward. c' illustrates how many times that the nodes are covered by the gateways (Actually, we delete the already covered nodes from the nodes set during each step, which means that the nodes can only be covered once. Thus, $c' = 1$). $\frac{e_m}{P_{max}}$ describes the ratio that the transmit power of the gateway that used by the nodes. Since we hope that the gateways can cover as many nodes as possible, the value of $\frac{e_m}{P_{max}}$ should be as less as possible. In each step, we also delete the already activated gateways from the gateway candidate set.

$$r_m = \begin{cases} -\alpha + \beta(1 - isA_m) \left(\sum_{1}^{n'} \frac{1}{c'+1} - \frac{e_m}{P_{max}} \right), & n' > 0; \\ 0, & n' = 0. \end{cases} \quad (17)$$

Based on the models above, we design H-DQN to solve the problem. Before the algorithm begins, we first initialize the environment which contains the information of the nodes and the gateways. Since the locations of the candidate gateways and the nodes are known, the power and distance distributions can be obtained after the initialization. When the algorithm begins, the network starts with a random process of the exploration by using the random probability ϵ for some episodes and records the empirical data in the replay buffer. Then, to avoid the model to fall into the local optimum, in the main training loop, we let the algorithm explore the solution based on greedy-based approach. After the environment is reset again, the network explores the solutions with a decaying epsilon during each episode. If the random probability is less than ϵ , the network selects an action randomly, while the random probability is larger than ϵ , the network selects the action with the maximum Q-value. During the step updating, we set a start power and a final power to update the rest power of the gateway unoccupied by the covered nodes. The gateway should cover nodes as many as possible to obtain larger reward. Based on the communication range and the power constraint, the algorithm deletes the already covered nodes and the power occupied by the nodes before next step begins. During each step, we first assume that this candidate location is deployed with a GW-X and calculate the reward obtained by executing this action. Then, we calculate the corresponding reward if we deploy a GW-Y at this location. After comparing these two rewards, the network should choose one type gateway to deploy and update the nodes information. When all the nodes are covered by the gateways or the system reaches the max step, the network outputs the reward in this episode. Actions, states and rewards in this episode would be added to the replay buffer for next few episodes. Then, the network calculates the loss by using gradient descent. After this, the algorithm runs the iteration loop to obtain more rewards and optimized results. The output is the number of the activated gateways and the average rewards. Algorithm 2 shows the iteration procedure of H-DQN.

VI. SYSTEM EVALUATION

In this section, we evaluate our system and algorithms by building an area of 100km×100km. The M candidate locations to deploy gateways and the locations of the different types of nodes are randomly selected in this area. Fig. 3 shows the diagram of the simulation configurations of 100 gateways and 100 nodes, the black stars illustrate the locations of the gateways, the circle dots show the locations of the SF-Ns and the triangle dots are the locations of the DF-Ns. After we have the locations of the candidate gateways and nodes, we can also obtain the power and distance distributions between the gateways and nodes. We treat the power and distance distributions as known parameters in $\mathcal{P}_2$.

Algorithm 2: Heuristic DQN (H-DQN) algorithm

Input: reset parameters, replay buffer
Output: activated gateways, average rewards, loss

- 1 Initialize the system parameters.
- 2 Put the random exploration results in the replay buffer.
- 3 **while** *Iteration < The maximum* **do**
- 4 **while** *Not all of the nodes are covered* **do**
- 5 Reset the total reward as 0;
- 6 Reset the reward as $-\alpha$;
- 7 Select an action based on ϵ -greedy;
- 8 Set an initial reward to activate the gateway;
- 9 Find the nodes within the range of the gateway
 the maximum transmit power that the gateway
 can sustain;
- 10 Find the maximum nodes the GW-X can cover
 and temporarily store the occupied power;
- 11 Find the maximum nodes the GW-Y can cover
 and temporarily store the occupied power;
- 12 Compare the reward of deploying GW-X and
 GW-Y, and select the larger one;
- 13 Obtain the reward in a single step and update
 the state;
- 14 Delete the covered nodes and the location of
 the gateway selected in last action.;
- 15 Obtain the new action and next state;
- 16 Put the action, state, reward, next state into the
 replay buffer;
- 17 Update the total reward;
- 18 **end**
- 19 Select some data from the replay buffer and
 calculate the loss between the online network and
 the target network;
- 20 Update the target network;
- 21 Iteration = Iteration + 1;
- 22 **end**
- 23 Return the number and the locations of the activated
 gateways, and the average reward of the network.

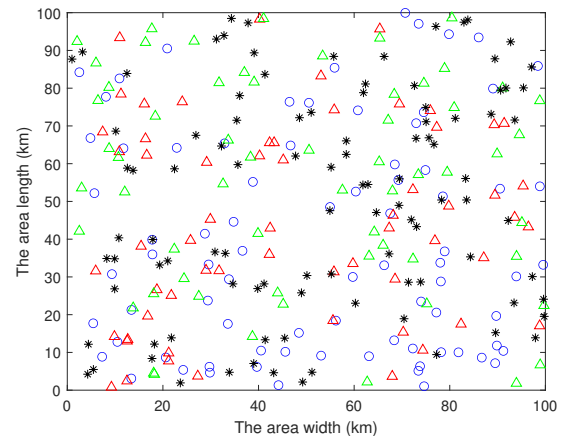


Fig. 3: Diagram of the simulation configurations of 100 gateways and 100 nodes.

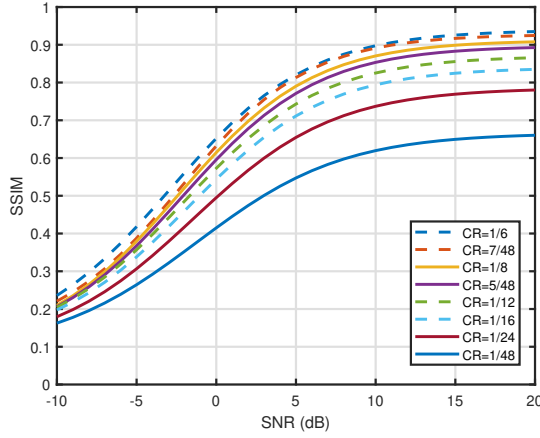


Fig. 4: SSIMs and the SNRs under different compression ratios (CR: compression ratio).

A. Parameter Settings

Fig. 4 shows the relation between SSIM and SNR under different compression ratios. SSIM increases as the SNR and compression ratio increases. Based on the data in [24], we can obtain the values of A_{1,c_k} , A_{2,c_k} , C_{1,c_k} and C_{2,c_k} by fitting the curve illustrated in Eq. (1). When the value of SSIM is fixed, it is noted that the smallest SNR can be obtained with $CR=1/6$.

After we obtain the power and distance distributions, we need to define some parameters in H-DQN. γ is the discount rate for computing the temporal difference. The buffer size `BUFFER_SIZE` is the maximum number of data transitions that should be stored in the replay buffer before overriding the old data transitions. The minimum replay size `MIN_REPLAY_SIZE` is the number of transitions that are supposed to replay in the replay buffer before we start training. The batch size `BATCH_SIZE` is the number of data transitions that we choose as the samples from the replay buffer when the algorithm operates the gradient descent. `EPSILON_START` is the start value of ϵ before it decays. `EPSILON_END` is the last value of ϵ after it decays. The target update frequency `TARGET_UPDATE_FREQ` is the number of steps that we need to update the parameters as same as the online parameters. `MAX_STEP` will let the algorithm stop in a single episode when the algorithm cannot find a reasonable solution. The learning rate `LEARNING_RATE` can be tuned to decide the duration of convergence. Table II shows the parameter settings in H-DQN. Since we need to maintain the reward within a reasonable range and make the procedure of convergence reasonably, we set α to 0.1 and β to 0.5 considering the normalization results of the rewards. Besides, we assume that 40 percents of the total nodes are SF-Ns, the residuals are DF-Ns.

B. Test Systems and Methods

After tuning the hyper-parameters and operating the network for large number of times, the convergence of reward and loss can be shown in Fig. 5. It is noted that after few steps, the

TABLE II: Parameter Settings

Parameters	Values
α	0.1
β	0.5
γ	0.95
<code>BUFFER_SIZE</code>	1000000
<code>MIN_REPLAY_SIZE</code>	1000
<code>BATCH_SIZE</code>	64
<code>EPSILON_START</code>	1
<code>EPSILON_END</code>	0.1
<code>TARGET_UPDATE_FREQ</code>	1000
<code>MAX_STEP</code>	100
<code>LEARNING_RATE</code>	0.001

average reward and the loss begin to converge gradually, and then become stable.

To investigate the performance of the proposed algorithm, we conduct some other algorithms based on the fixed map which is generated by using the same random seed. We adopt the fixed diagram with an area of $100\text{km} \times 100\text{km}$ and also deploy the two types of nodes by using the same seed in different algorithms. We conduct BB, GBA, Q-learning algorithm and H-DQN to compare the performance. Firstly, since the performance of the network is closely related to the number of nodes, we try to increase the number of nodes and evaluate the impacts on the four algorithms. Then, we investigate the operation time of the four algorithms. Besides, since we build a new heterogeneous network for SemCom application, we compare the single coverage problem and our approach to discuss the performance of the new proposed architecture.

C. Impacts of the Number of Nodes

Based on the $100\text{km} \times 100\text{km}$ diagram, we investigate the relation between the number of nodes and the performance of the algorithms. We increase the number of nodes based on this diagram and observe the number of gateways that need to be deployed at the candidate locations. Fig. 6 shows the average, minimum, maximum, and standard deviation of the number of needed gateways when the number of nodes increases. As the number of nodes increases, the result of needed gateways also increases. Fig. 7 illustrates the impact of different number of nodes on the performance. Based on the same diagram, BB and GBA have fixed results in every iteration, shown as the orange line and the green line, while Q-learning and H-DQN show the average results after every episode, shown as the blue line and the red line. This is the reason why we obtain nonintegral results from Q-learning and H-DQN. When there is a small quantity of nodes in the network, the difference between H-DQN and BB is relatively large. While the number of nodes gradually increases, the difference between H-DQN and BB reduces. Besides, H-DQN has better performance than Q-learning and GBA. Considering the real scenarios when deploying gateways, H-DQN has reasonable results.

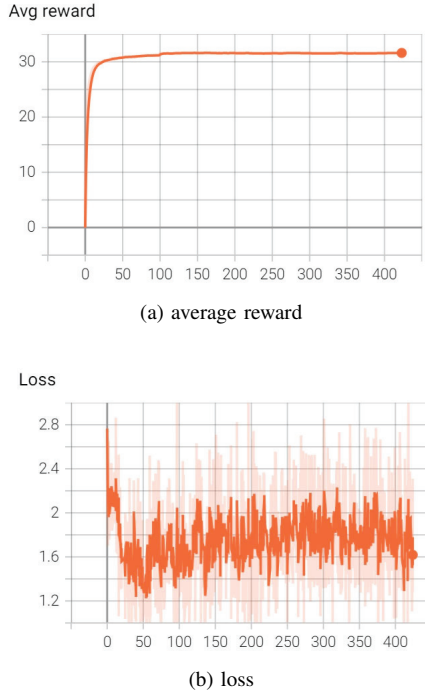


Fig. 5: The convergence of H-DQN algorithm: (a) average reward; (b) loss.

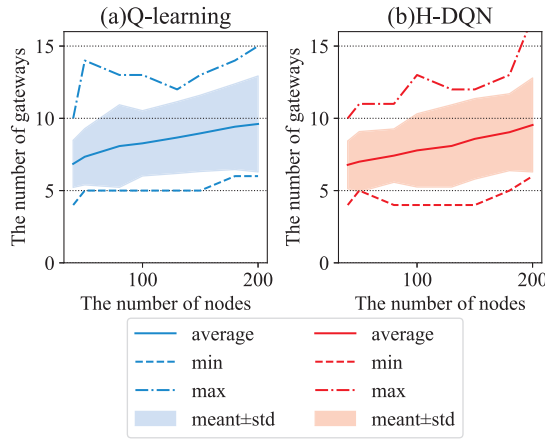


Fig. 6: The number of gateways of Q-learning and H-DQN.

D. Operation Time Analysis

Based on the same diagram and increasing number of nodes, we also investigate the operation time of the four algorithms. Fig. 8 shows the operation time when the number of nodes increases. In Fig. 8(a), we compare the four algorithms, and results show that BB has larger iteration time than the other three algorithms. Then, we compare the GBA, Q-learning and H-DQN in Fig. 8(b), which illustrates that H-DQN has the minimum operation time. Fig. 9 shows the average, minimum, maximum and standard deviation of the operation time of the four algorithms. Q-learning and H-DQN perform better than GBA and BB, while H-DQN saves more than 50% operation time when compared to Q-learning and over 20× operation time than BB. From the experimental results above,

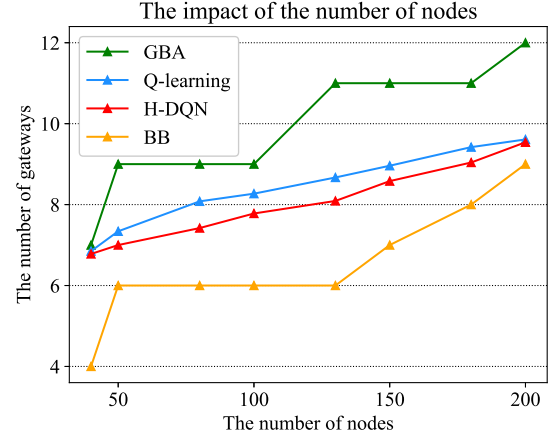


Fig. 7: The number of gateways of the four algorithms.

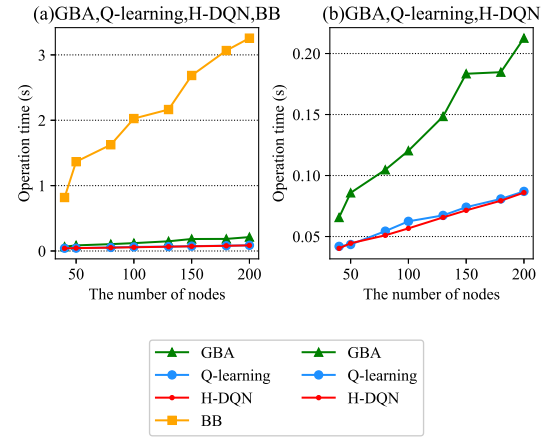


Fig. 8: Operation time of the four algorithms.

though H-DQN cannot provide the optimal solutions, it has the optimal iteration time for each solution compared to the other algorithms. Due to the physical map and the real deployment plan, H-DQN provides faster results and solutions, which can improve the efficiency of deployment for industrial applications.

E. Single Coverage System Analysis

In this subsection, we compare the single coverage system with our proposed architecture. In single coverage system, once the nodes are covered by one gateway, they can communicate with the gateway and maintain the communication quality, which means that the nodes have consistent requirements. We also adopt H-DQN to evaluate the single coverage system, and the results are shown in Fig. 10. It is noted that in Fig. 10(a), the number of gateways that needed in our system is less than that in the single coverage system, especially when the number of nodes in the network is large. Therefore, our system consumes limited resources and satisfies the different requirements of the heterogeneous nodes compared with the single coverage system. Besides, we investigate the system throughput of the two systems by applying the SINR of the

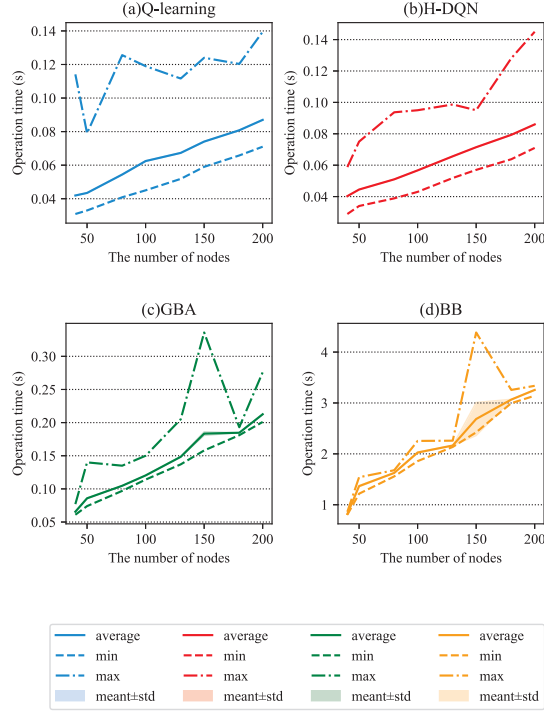


Fig. 9: The operation time.

nodes used in [7], shown in Fig. 10(b). The throughput of our system is $1.3\times$ larger than that of the single coverage system with less gateways, which notes that our system adopts less gateway resources and obtains higher system performance.

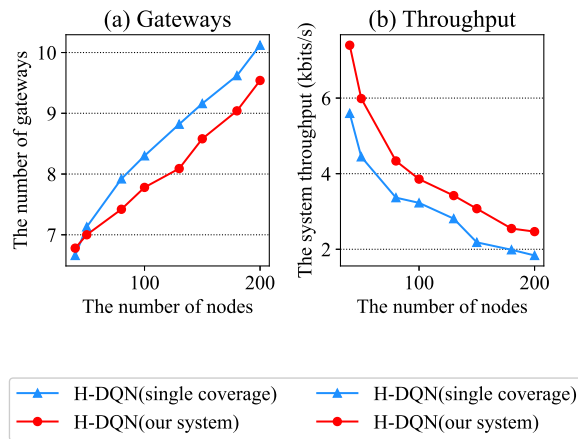


Fig. 10: The number of gateways and the system throughput for our system and the single coverage system.

F. Discuss

From the experimental results above, our proposed architecture can obtain better performance than the single coverage system with very limited number of gateways. Furthermore, our designed H-DQN algorithm outperforms the other algorithms which includes GBA and Q-learning, and saves more

operation time than BB. Though our system and algorithm can be reasonably adopted in real application for SemCom, there are still some crucial contents to be discussed.

Improving the Robustness: H-DQN is designed based on DQN, which means that our system can be relatively adaptable in various environments. However, when applied in industrial scenarios, the system suffers more severe metal interference. The parameters in this model may be retuned to adapt the dynamically changing environment. Therefore, we are supposed to make the system and the algorithm more robust in the future. By doing this, we may introduce some prediction models to monitor the dynamic changes and make the determination to adaptively adjust the model parameters. Besides, the replay buffer and the reward function should be further improved to balance the exploration of the models.

Enriching the dataset: Our system proposes a new architecture for SemCom adopted in industrial applications. However, the real dataset for this kind of transmission in industry is very limited. More industrial data should be collected and corresponding dataset should be built for semantic transmission in the future. By doing so, some industrial nodes and gateways are supposed to be deployed in real factories or production lines. Our proposed architecture can be a novel approach to solve the coverage problem and generate the dataset. However, data collected from industry may suffer from severe metal interference and can be sensitive to the dynamic industrial environment. Therefore, the data analysis and processing can be crucial in SemCom architecture. Furthermore, based on the various datasets from industry, our model can be further significantly improved to accelerate the training process.

Optimizing the resource: As the dataset will be gradually abundant in the future, the retraining of the model may be transferred to edge servers. More lightweight models are supposed to be designed for industrial nodes. Therefore, in order to ensure the transmission performance and improve the energy efficiency, the trade-off between the arrangement of the calculation resources in the nodes and the servers should be discussed. Besides, when applied in real industrial scenarios, large number of nodes operate based on contentions, which would cause severe conflicts and impair the efficiency. Therefore, contention resolution is also supposed to be a promising research area in the future.

Protecting the privacy: Since SemCom networks transmit the semantics instead of the raw data, attackers may transmit the same semantics which have totally different raw data to replace the transmitted data. Therefore, the security and privacy concerns have an impact on SemCom's efficiency and accuracy. In the future, Blockchain with cryptography may help protect the data transmission, leveraging its decentralized, secure, and transparent mechanisms. Blockchain can ensure that only authorized parties can access and interpret the meaning behind the semantic data, and reduce the risk of single-node failures and enhance the resilience of SemCom systems.

VII. CONCLUSION

In this article, we proposed a new heterogeneous architecture for SemCom in industrial networks. Unlike prior work

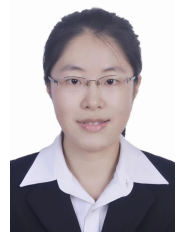
on SemCom or IoT networks, we focused more on the fragmentation of the industrial application scenarios. In order to meet the heterogeneous requirements of the nodes, we built a hybrid structure of gateways to cover the nodes. Considering the constraints in SemCom, we formulated the problem as an NP-hard problem and simplified it by comprehensive analysis. We designed GBA and H-DQN to solve the problem. From the experimental results, the proposed H-DQN required less gateways than the GBA and Q-learning algorithms, and saved more operation time than the BB, GBA and Q-learning algorithms. Furthermore, our system outperformed the single coverage system with higher system throughput by deploying limited gateways. In the future, the model should be more robust to adapt to the dynamic environment, and the dataset of industrial networks should be enriched to help train the models. Besides, the balance between the transmission performance and the energy efficiency should be further investigated.

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Yuqi Sun (Member, IEEE) received the B.Sc. degree in Information Science and Engineering from Central South University, Changsha, China, in 2016 and the Ph.D. degree in Control Science and Engineering from Zhejiang University, Hangzhou, China, in 2021. She is currently a Lecturer with the School of Information Science and Technology, Hangzhou Normal University, Hangzhou, China. Her current research interests include Low-Power Internet of Things, Industrial Internet of Things, and wireless communication.



Fangyuan Xing (Member, IEEE) received the B.Eng., M.Eng., and Ph.D. degrees from Dalian University of Technology, Dalian, China, in 2014, 2016, and 2020, respectively. From 2018 to 2019, she was a Visiting Ph.D. Student with University of British Columbia, Vancouver, BC, Canada. From 2020 to 2022, she was a Postdoctoral Fellow with Zhejiang University, Hangzhou, China. Since November 2022, she has been an Associate Professor with the School of Cyber Science and Engineering, Southeast University, Nanjing, China. Her research interests

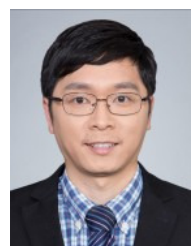
include wireless communications and Internet of Things.



Zhenwei Guo received the B.E. and Ph.D. degrees in automation from Zhejiang University, Hangzhou, China, in 2016 and 2021, respectively. He is currently an associate research fellow with the Research Center of Cyber Science and Technology, Hangzhou Innovation Institute of Beihang University. His research interests include P2P energy trading, blockchain technology, privacy computing and distributed optimization.



Huimin Chen received the B.Sc. degree from the College of Automation Engineering, Nanjing University of Aeronautics and Astronautics, Nanjing, China, in 2018. She is currently pursuing the Ph.D. degree with the Department of Control Science and Engineering, Zhejiang University, Hangzhou, China.



Shibo He (Senior Member, IEEE) received the Ph.D. degree in control science and engineering from Zhejiang University, Hangzhou, China, in 2012. He is currently a Professor with Zhejiang University. He was an Associate Research Scientist from March 2014 to May 2014, and a Postdoctoral Scholar from May 2012 to February 2014, with Arizona State University, Tempe, AZ, USA. From November 2010 to November 2011, he was a Visiting Scholar with the University of Waterloo, Waterloo, ON, Canada. His research interests include Internet of Things,

crowdsensing, big data analysis, etc.